

Rare Nuclei on PanNuke Are a Detection Problem: Diagnosis, Two Failed Data-Side Fixes, and a Counterfactual Stress-Test Evaluation

Stage Report I (Course Project bdd26)

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Abstract

On PanNuke, swapping backbones moves mean panoptic quality (mPQ) by only 1–2 points, while the *Dead* (apoptotic) class stagnates at $PQ \approx 0.13$ – 0.18 for every published model. This stage report asks why, and whether recent “new paradigms” (pathology vision–language models and controllable generative models) can address either the rare-class deficit or robustness evaluation. We first reproduce two baselines under the official 3-fold protocol (HoVer-Net 0.456 mPQ, CellViT-UNI 0.500) and add a third evaluator, HoVer-NeXt-T (0.458), after diagnosing that our port’s original 0.358 was a decoding artifact, not a model gap. Second, we show the Dead deficit is concentrated in *detection*, not classification: 38.5% of Dead ground-truth nuclei receive no overlapping prediction, and inside those missed nuclei the foreground head is silent (mean nuclear-probability 0.16) while the type head assigns $P(\text{Dead}) \approx 0.26$. Third, under a pre-registered multi-seed protocol, five interventions are null or harmful: post-hoc re-typing with a pathology VLM (CONCH), training-free missed-nucleus recovery, class-weighted and size-weighted foreground losses, and, on the data side, failure-driven copy-paste (Dead PQ -0.031) and in-context diffusion synthesis with self-consistent labels (mPQ unchanged). Fourth, we build a three-architecture $\times$ three-dataset zero-shot cross-domain benchmark (CoNIC, MoNuSAC, PUMA) and *PanNuke-CF*, a counterfactual test set that re-renders fixed PanNuke labels under interventions on tissue context and stain. Counterfactual stress *inverts* the real cross-domain ranking (Spearman -0.89 across 6 evaluator points; -1.00 on architecture means): an axis decomposition shows real between-architecture transfer differences live almost entirely on the detection axis, which label-fixed rendering cannot stress by construction, while on the typing axis, which it does stress, counterfactual ranking agrees with real transfer. We conclude counterfactual rendering is a valid *typing* stress test but not a transfer surrogate, and that the Dead detection deficit is architectural: it is robust to every loss-level and data-level intervention we tested, and follows the models out-of-domain (mean Dead PQ 0.002–0.033 on PUMA, vs 0.13–0.18 in domain).

1 Introduction

Nuclei instance segmentation and classification is a standard workload in computational pathology, and PanNuke [1] its standard benchmark: 205,000 annotated nuclei across 19 tissues, evaluated by tissue-averaged mean panoptic quality (mPQ). The published leaderboard is lopsided: five years of architectural progress compress into ~ 1 – 2 mPQ points (HoVer-Net [2] 0.463 $\rightarrow$ CellViT-SAM-H [3] 0.498 $\rightarrow$ best current models ≈ 0.51), while the *Dead* (apoptotic bodies) class stays at PQ 0.13–0.18 and touching-nucleus merge/split errors persist. Two recent paradigm families promise

more than backbone engineering: pathology vision–language models (VLMs) [8] as priors for type classification, and controllable generative models [10] as “tissue simulators” that can synthesise targeted training data or re-render evaluation images under interventions.

This project was organised as three pre-registered pillars with explicit go/no-go rules, on the logic that negative results are only useful if their acceptance criteria are fixed in advance: **A** — language-anchored typing (attack rare-class *classification* with the CONCH text space); **B** — failure-driven synthesis (attack Dead/small-nucleus *detection* with generated data); **C** — evaluation (attack robustness measurement with real cross-domain transfer plus a counterfactual test set). All three pillars have now closed; this report documents what was found. Its contributions:

1. **Reproduction under a single strict protocol** (Sec. 3): HoVer-Net and CellViT-UNI trained by us and evaluated identically (last checkpoint, validation-tuned decodes); HoVer-NeXt-T runs the official released *per-fold* weights (their val-selected checkpoints; the fold mapping is verified to match ours exactly) under the same validation-only decode tuning. Along the way we quantify how much a released evaluation recipe (16 stochastic colour-TTA views + thresholds tuned on the *test* fold) inflates reported numbers (Sec. 3.1).
2. **A decomposition of the Dead deficit** (Sec. 4.5): it is a *detection* failure concentrated in small and isolated nuclei, invisible to the foreground head but partially “seen” by the type head—which is exactly why every typing-side intervention fails.
3. **Pre-registered negative results** (Secs. 4, 5): CONCH re-typing, training-free recovery, loss re-weighting (3 seeds), failure-driven copy-paste, and in-context diffusion synthesis are all null or harmful, with seed-noise calibration that says which effect sizes would have been believable.
4. **A 3×3 architecture × dataset zero-shot cross-domain benchmark** (Sec. 6.1), including PUMA as the only sizable external source of apoptotic-body Dead labels we could find, and a conversion/verification pipeline for CoNIC and MoNuSAC.
5. **PanNuke-CF**, a label-fixed counterfactual rendering benchmark (Sec. 6.2), with a validity analysis that reaches a mechanistic verdict: counterfactual appearance stress cannot probe detection (by construction) and therefore *inverts* the real transfer ranking at the mPQ level, while remaining valid as a typing stress test.

Everything is one codebase; every number below comes from the same evaluator, which was checked bit-exact against the official PanNuke-metrics implementation (whose published `class_stats.csv` has a column-swap bug we document in the repository).

2 Related Work

PanNuke models. HoVer-Net [2] introduced the horizontal–vertical watershed decode still used (in variants) by most successors. CPP-Net [6], HoVer-NeXt [5], CellViT [3] and its successors (LKCell, CellVTA, CellViT++ [4]), and PromptNucSeg [7] populate the 0.46–0.51 mPQ band. Publicly released per-fold weights exist for only some of these; several official checkpoints (e.g. HoVer-Net’s) were trained on all folds and can only serve as pipeline sanity checks, never as test results—we enforce this distinction throughout.

Pathology foundation models and VLMs. Frozen pathology foundation models do not automatically beat ImageNet CNNs on dense tasks [9], and the pathology VLM CONCH [8] is on par with pure-visual models for dense prediction while providing an aligned text space. Pure prompt-based interfaces fail badly on nuclei (SAM3-style prompting achieves $< 10\%$ mIoU on PanNuke-type tasks), and the closest prior work (PromptNu, MONCH) either lacks an official PanNuke configuration or reports only semantic segmentation. We found no prior work combining a pathology VLM text space with instance segmentation under the official PanNuke protocol.

Generative augmentation and stress tests. Synthesis for nuclei segmentation has been reported to help *rare classes* (DiffMix, HistoSmith, NucleiMix [11]) but never, to our knowledge, under the official 3-fold PanNuke protocol with mPQ; existing studies use non-standard splits and Dice/AJI/F1. Counterfactual stress testing of imaging models has been demonstrated in radiology [12]; we are not aware of an instance segmentation analogue. PixCell [10] (with its CellControlNet) is the strongest openly available controllable nuclei generator, and is the base of both our synthesis experiments and our counterfactual renderer.

3 Experimental Setup

Data and protocol. PanNuke folds 1–3 with the official assignment (train/val/test = 1/2/3, 2/1/3, 3/2/1); all numbers are 3-split means with no test-based early stopping: last checkpoint for our trained HoVer-Net/CellViT-UNI, the official released per-fold (val-selected) checkpoints for HoVer-NeXt-T. External datasets (Sec. 6.1) are zero-shot: class mappings to the 5 PanNuke classes are fixed per dataset, absent classes are excluded from mPQ per the official protocol.

Metrics. Official tissue-averaged mPQ/bPQ (bit-exact reimplementations of PanNuke-metrics), strict mPQ (FPs on classes absent from an image are penalised rather than ignored), AJI/AJI⁺, detection F_d and classification F_c (12 px centroid matching), and an IoU-0.5 error decomposition (matched / merged / missed-no-overlap / missed-poor-shape / split, per class). TTA (8 dihedral views with HV-channel remapping) is reported separately.

Noise calibration (used for every verdict). Across 3 seeds of the identical CellViT-UNI recipe: seed std is 0.0021 mPQ but **0.0074 Dead PQ**, and the image-bootstrap CI half-width on Dead is 0.02–0.03. Rules we enforce: single-seed Dead changes below ~ 0.01 – 0.015 are noise; every intervention verdict is a 3-seed (or multi-fold) statement with paired, tissue-stratified image-bootstrap 95% CIs; validation-fold decisions only.

3.1 Baselines

Table 1: both classic baselines reproduce (HoVer-Net slightly below paper, consistent with common reproductions; CellViT-UNI slightly above). HoVer-NeXt-T required porting the official released per-fold weights and initially scored 0.358 mPQ; ablations showed the *predicted maps were fine* and the loss was entirely in the decode. Its released 0.477 recipe averages 16 stochastic spatial+colour TTA views and tunes per-class thresholds *on the test fold*. We instead use its own validation decode (flat fg/seed thresholds) tuned on each split’s *validation* fold—the same protocol as our other baselines—yielding 0.4579. The gap (0.458 vs 0.477) is a measure of what test-side tuning + stochastic colour TTA buy, and we flag that recipe differences of this size (0.02 mPQ) are larger than most leaderboard gaps. HoVer-NeXt-T earns its place as our third evaluator through a *different*

Table 1: Baselines under the official 3-fold protocol (3-split mean; no TTA, TTA in brackets). Last checkpoint for HoVer-Net/CellViT-UNI (our training); official released per-fold weights for HoVer-NeXt-T. “Published” = paper-reported numbers under their own recipes.

Model	mPQ	bPQ	Dead PQ
HoVer-Net (ours)	.4564 [.4666]	.6613 [.6714]	.127 [.128]
published	.463	.660	.139
CellViT-UNI (ours)	.4995 [.5047]	.6654 [.6706]	.176 [.177]
published (CellViT++)	.492	.664	.154
HoVer-NeXt-T (ours)	.4579	.6367	.136
published	.477	.656	.154

Table 2: Post-hoc re-typing on CellViT-UNI, split 1 (test fold 3).

Method	val mPQ	test mPQ	test Dead PQ
seg vote (pipeline)	.4835	.4951	.142
+ class-bias recalibration	.4873	.4953	.135
× CONCH zero-shot ($\alpha=.1$)	.4835	.4938	.141
× CONCH linear ($\alpha=.3$)	.4836	.4920	.141
CONCH linear alone	.3645	.3821	.145

error profile: its flat decode merges about twice as many touching nuclei as the watershed decodes (merged GT rate 0.096–0.101, vs 0.044 for CellViT-UNI and 0.049 for HoVer-Net), and it misses more background nuclei than CellViT-UNI (0.122–0.131 vs 0.102–0.109).

4 Pillar A: The Dead Deficit Is a Detection Deficit

4.1 CONCH semantics live at the context scale (feasibility)

Zero-shot CONCH classification of ground-truth nuclei: tight crops are near chance (32 px: 0.27), full-context patches with radius-restricted attentional pooling reach balanced accuracy 0.695 (R=32 px, one forward pass per patch), with LLM morphology descriptions beating bare class names by 8–11 points. Dead is the *most* text-recognisable class (recall 0.88) yet the worst class for segmentation models—but at natural prevalence its zero-shot precision is ≈ 0.10 , so it can only be a complementary signal. Text-anchored initialisation of a linear head helps only in the few-shot regime (≤ 25 nuclei/class); with full PanNuke labels it is irrelevant. This redirected pillar A from classification to detection.

4.2 In-pipeline re-typing: null

Fusing CONCH (zero-shot or linear-probe) into the type decision of CellViT-UNI, with all knobs tuned on the validation fold (Table 2): no configuration beats the pipeline’s own seg-vote typing, and the recalibration control matches the best fusion. With TTA, validation tuning picks $\alpha = 0$ for both fusions.

4.3 Training-free recovery: null

A 96-configuration sweep over foreground mixing, dead-prior boosting, blob thresholds and orphan recovery on the raw NP/HV/TP maps (3 splits; selection on validation): every family lands within

Table 3: Error decomposition of CellViT-UNI on fold 3 (IoU 0.5; fraction of GT nuclei).

GT class	matched	merged	missed	miss (shape)	split	type matched
Neoplastic	.811	.048	.092	.040	.009	.905
Inflammatory	.849	.046	.079	.023	.004	.817
Connective	.752	.038	.153	.044	.013	.794
Dead	.470	.089	.385	.048	.008	.690
Epithelial	.837	.042	.075	.036	.011	.930

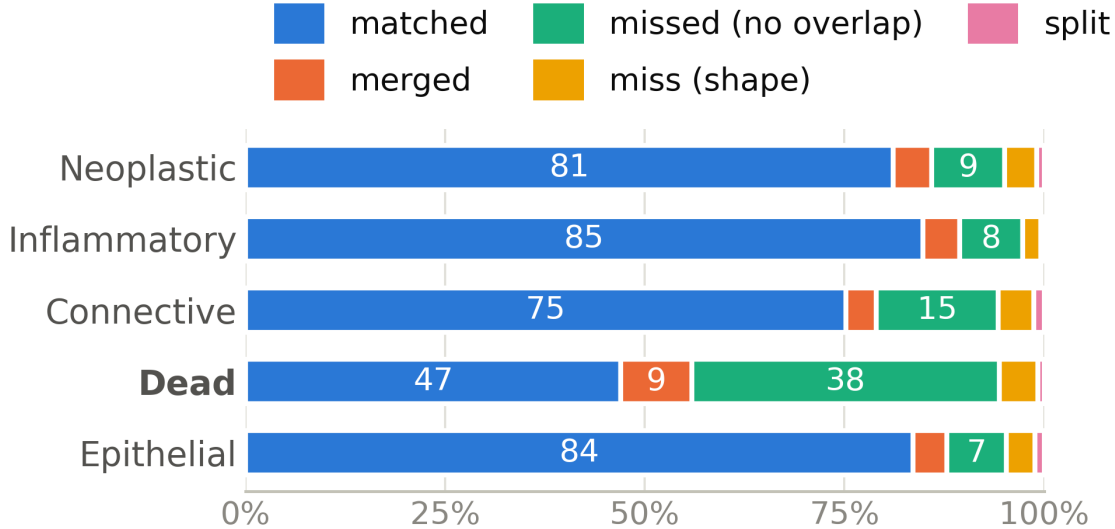


Figure 1: Error decomposition of CellViT-UNI on fold 3 (IoU 0.5; share of GT nuclei per class), visualising Table 3: the Dead row is dominated by missed detections (38.5% of Dead nuclei have no overlapping prediction; other classes 8–15%), not by typing errors.

± 0.0004 test mPQ of the official config, and the threshold-only control does as well as any recovery variant. The missed Dead nuclei are not recoverable from this model’s outputs.

4.4 Loss-side ablations: null or a trade

Pixel-weighted foreground CE on split 1 with 3-seed baselines: Dead-pixel weight $\times 11$ (M1) null; fully balanced sampler (C1) null; small-nucleus weight $\times 6$ (M2) looked promising in one seed ($+0.0042$ test mPQ, CI excluding 0) but across 3 seeds is mPQ -0.0010 $[-0.0043, +0.0022]$ with a *significant* bPQ loss -0.0024 $[-0.0048, -0.0002]$ and a non-significant Dead gain $+0.0108$ $[-0.0021, +0.0238]$ —a trade, not a contribution, and a clean demonstration that single-seed mPQ effects below ~ 0.004 on this benchmark are seed luck.

4.5 Diagnosis: where Dead performance is lost

Table 3: 38.5% of Dead nuclei have *no overlapping prediction at all* (other classes 8–15%), while Dead type-when-matched is 0.69—the loss is detection-first. Inside the missed Dead nuclei the foreground head is silent (mean NP probability 0.164; fires in 17% of them) yet the type branch assigns $P(\text{Dead}) \approx 0.26$ (vs ≈ 0.01 for missed nuclei of other classes): the network “sees” dead

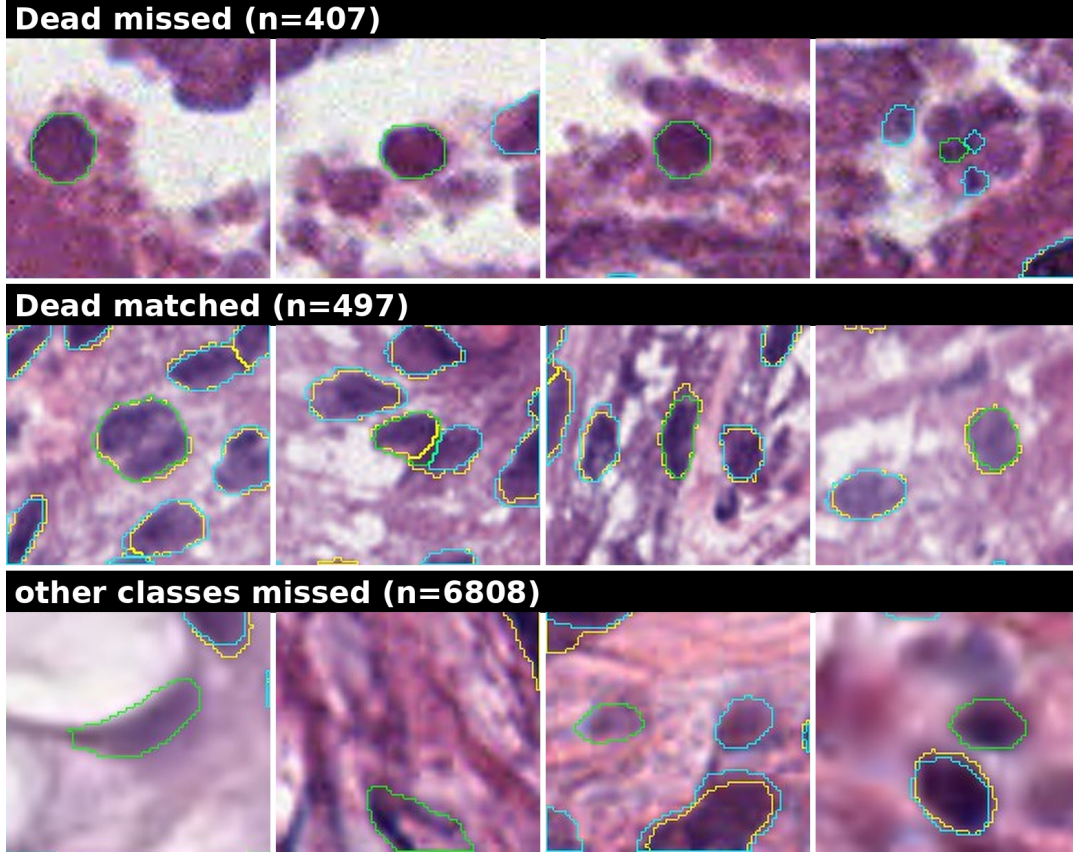


Figure 2: Dead ground truth (green) vs. predictions (yellow) for CellViT-UNI on fold 3 (cyan: other ground truth); four crops per group. Top “Dead missed”: no prediction corresponds to the green outline. Middle “Dead matched”: recovered nuclei. Bottom “other classes missed”: misses are far rarer off the Dead class. Missed Dead are tiny dark fragments and chromatin-poor apoptotic bodies that carry no foreground response.

material that the foreground head drops. Size drives much of it (Dead median area 107 px vs 368 for other classes; 24% of Dead are <60 px vs 6%), but an appearance effect survives size stratification: large Dead (>200 px) are missed $6\times$ more often than large nuclei of other classes (0.214 vs 0.035). Qualitatively (Fig. 2), missed Dead are karyorrhectic fragments, chromatin-poor eosinophilic bodies, and ambiguous annotations. Conclusion: fix detection or the data—which defines pillar B.

5 Pillar B: Failure-Driven Synthesis Does Not Fix Detection Either

5.1 What failure layouts look like

Out-of-fold failure mining (split-2 model on fold 1; identical patterns on fold 3 test): missed Dead are *isolated*—Dead with zero touching neighbours have miss rate 0.39–0.42 ($\sim 90\%$ of all Dead), and missed Dead touch another nucleus in $<5\%$ of cases. Merges are an intra-class cluster phenomenon (73–82% of merged Dead touch another Dead: karyorrhexis fragments). Cross-class touching, the

intuitive hypothesis, is rare and not the failure mode. The originally planned “Dead glued to Neoplastic” intervention was therefore replaced by (1) isolated small-nucleus insertion into empty stroma and (2) dense same-class clusters.

5.2 CP1: failure-driven copy-paste is harmful

Donor nuclei (50–400 px, Dead-weighted 0.4) pasted into nucleus-free stroma with 8 px clearance *before* geometric/photometric augmentation (so pasted nuclei are augmented identically; a v1 shortcut caught by visual review and fixed, alongside border-fragment and unannotated-nucleus collisions). Result vs 3-seed baseline (split 1; test fold 3): $\Delta\text{mPQ} +0.0009$ $[-0.0024, +0.0043]$ but $\Delta\text{Dead} -0.0310$ $[-0.0576, -0.0100]$ (TTA-consistent). The damage is in *typing*, not detection: matched-Dead type accuracy .690→.609, confusion increases in both directions. The validation fold, however, pointed the *opposite* way on Dead (+0.0235): Dead PQ swings ~ 0.05 between folds, the third independent confirmation of the noise rules. Interpretation: transplanting real appearances into new contexts teaches “small isolated nucleus in clean stroma” as a Dead-ish cue and blurs the type boundary.

5.3 In-context synthesis (PixCell): from an appearance gap to a structural blocker

PixCell-256 + Cell-ControlNet with real fold-1 masks as condition: placement is faithful ($\sim 95\%$ of mask objects rendered at the right place/size, median IoU 0.68) but the $20\times$ generator has a structural appearance gap on $40\times$ PanNuke (flat chromatin, -24% high-frequency content, size-biased under-rendering of large masks). LoRA adaptation (two rounds, r16 and r8 with snapshot selection) made every image-space metric *worse* and was abandoned; a Reinhard LAB transfer to the paired real patch closes the colour gap (OD-L1 $0.072 \rightarrow 0.009$). With the locked recipe (base model + paired context + Reinhard), visual review of 32-image smokes passes (colour within $\sim 3/255$ of paired real), but review exposed a structural blocker: the ControlNet condition is a *binary* mask and the context embedding carries no class information, so the generator cannot render class-targeted appearance—0 of 729 kept objects were teacher-typed Dead ($P(\text{Dead}) \leq 0.005$). Forcing donor-class labels would recreate exactly the CP1 failure (labels that contradict appearance). SYN1 therefore used *class-agnostic*, *teacher-self-consistent* labels, which can only help through detection: 1146 synthetic images at 50% mixing, exact base recipe otherwise. Result (test fold 3): mPQ 0.4900 vs base 0.4951 (bottom of the 3-seed range 0.4910–0.4951), Dead 0.1521 vs 0.1424 ($\approx +1\text{pt}$, at the edge of seed noise, with $\text{PQ}^+/\text{DQ}^+/\text{SQ}^+$ all flat: the sliver of Dead gain is better type pairing, not better segmentation) and Inflammatory -1.0 – 1.6pts . Per the pre-registered rule: null.

5.4 Pillar B verdict

The Dead/small-nucleus detection deficit is not fixable by the data-side levers available without retraining the generator’s conditioning: real-appearance transplant (CP1) actively harms Dead typing, and in-context synthesis with self-consistent labels (SYN1) is null. Together with pillar A’s nulls (re-typing, recovery, loss re-weighting), the deficit looks architectural.

Table 4: Zero-shot cross-domain (3-split mean, no TTA). In-domain mPQ: CellViT 0.4995, HoVer-Net 0.4564, HoVer-NeXt-T 0.4579.

Model	Dataset	mPQ	bPQ	F_d
CellViT-UNI	CoNIC	.3351 $\pm$.0037	.5314	.7816
	MoNuSAC	.2495 $\pm$.0218	.5616	.7652
	PUMA	.4518 $\pm$.0031	.7177	.8831
HoVer-Net	CoNIC	.2610 $\pm$.0061	.5136	.7669
	MoNuSAC	.0370 $\pm$.0003	.2756	.3658
	PUMA	.3203 $\pm$.0139	.7136	.8714
HoVer-NeXt-T	CoNIC	.2786	.4811	.725
	MoNuSAC	.1123	.4025	.628
	PUMA	.3813	.6888	.848

6 Pillar C: Cross-Domain and Counterfactual Evaluation

6.1 Real cross-domain zero-shot: 3 $\times$ 3

External sets, converted and montage-review-verified: **CoNIC** (19,476 tiles; the 112 patches sourced from PanNuke excluded; 20 $\times$ $\rightarrow$ 2 $\times$ upsampled; classes Inflam/Epith/Conn), **MoNuSAC** test (443 tiles; Inflam/Epith), **PUMA** melanoma (3,278 tiles; all five PanNuke classes present, including 1,991 apoptotic $\rightarrow$ Dead nuclei—the only sizable external Dead supply we found; GT is HoVer-Net-initialised and pathologist-corrected, so absolute recall has a ceiling that applies equally to all models).

Table 4: all models degrade, but differently. HoVer-Net nearly *collapses* on MoNuSAC (F_d 0.37; per-split 0.26–0.50): a detection failure. On PUMA no model collapses in detection (F_d 0.85–0.88)—the loss there is *typing* (strict mPQ 0.31 vs mPQ 0.38 for HoVer-NeXt-T), and **mean Dead PQ falls to 0.002 (CellViT-UNI), 0.005 (HoVer-NeXt-T) and 0.033 (HoVer-Net)** despite 1,991 apoptotic GT nuclei (in domain: 0.176, 0.136, 0.127): the Dead failure mode follows the models out-of-domain. TTA does not rescue transfer (+0.005 CoNIC, -0.001 MoNuSAC). Real domain drop (in-domain minus mean external mPQ) ranks architectures CellViT 0.154 < HoVer-NeXt-T 0.199 < HoVer-Net 0.246, consistently on every dataset.

6.2 PanNuke-CF: label-fixed counterfactual rendering

500 fold-3 patches with *fixed GT labels* are re-rendered by the (frozen) generator under a factorial appearance intervention—context {paired, swapped-tissue} $\times$ Reinhard target {self, donor}—giving arms **control/stain/ctx/tissue**, plus the untouched **real** images and 3 generator-seed replicates of control for a noise floor. Generator noise sits an order of magnitude below the intervention effects (control-replicate mPQ std 0.002–0.006). Two calibration facts: generated **control** images are *easier to segment* than real (bPQ 0.78 vs 0.67) but carry *type-inconsistent* appearance (strict mPQ 0.36 vs 0.45)—so CF comparisons are always arm-vs-control, never arm-vs-real; and blank patches must be excluded (the context embedding is a content channel).

Context swap is a strong stressor (Epithelial PQ collapses by 0.23–0.43 while F_d drops only 0.08–0.10: it destroys *type* information), stain a mild one. The validation question is whether these counterfactual drops predict *real* cross-domain drops across evaluators.

Table 5: PanNuke-CF, architecture means over 2 splits (mPQ; paired Δ vs control mean in brackets, image-bootstrap 95% CI all exclude 0 except HoVer-NeXt-T stain).

Architecture	real	control	stain	ctx	tissue
CellViT-UNI	.495	.449	.426 [−.023]	.259 [−.190]	.258 [−.191]
HoVer-Net	.449	.440	.373 [−.066]	.300 [−.140]	.293 [−.146]
HoVer-NeXt-T	.456	.425	.417 [−.008]	.262 [−.163]	.260 [−.164]

Table 6: Axis decomposition (architecture means): each drop split into detection (F_d) and typing (bPQ–mPQ gap increase). Real between-architecture differences live on the detection axis, where CF stress is architecturally homogeneous.

Architecture	Real (in-domain – ext)			CF ctx (control – ctx)		
	mPQ	F_d	typing	mPQ	F_d	typing
CellViT-UNI	.154	.007	+0.088	.190	.102	+0.082
HoVer-Net	.246	.163	+0.069	.140	.084	+0.034
HoVer-NeXt-T	.199	.045	+0.085	.163	.090	+0.064

Axis-matched rank agreement (all 6 points / architecture means):
mPQ: −0.89 ($p=.033$) / −1.00; detection F_d : −0.83 ($p=.058$) / −1.00; typing: +0.71 / **+1.00**.

6.3 Validity: counterfactual stress inverts the real ranking

Across 6 evaluator points (3 architectures $\times$ 2 splits), CF drop vs. real domain drop: Spearman $\rho = -0.89$ for **ctx** (exact permutation $p = 0.033$), -0.89 for **tissue**, $+0.43$ n.s. for **stain**. On architecture means (the honest test; the two splits of one architecture are near-duplicates) both ctx and tissue give $\rho = -1.00$: a perfect inversion—the model most damaged by context swap in-domain (CellViT) is the most robust real-domain transferrer, and vice versa.

Table 6 gives the mechanism. Real transfer differences between architectures concentrate on the *detection* axis (CellViT loses 0.007 F_d ; HoVer-Net loses 0.163—its MoNuSAC collapse), while typing-gap increase is similar for all (0.07–0.09). But label-fixed rendering *cannot* stress detection: the generator re-renders well-formed nuclei at fixed positions, and indeed CF F_d drops are homogeneous (0.08–0.10) across architectures with no ordering information. Where CF *can* apply stress—typing under context/stain mismatch—its ranking agrees with real typing transfer ($\rho = +1.00$ on architecture means; $+0.71/+0.83$ over all 6 points). Caveats: $n = 3$ architectures (an arch-mean $|\rho| = 1$ has exact $p = 1/3$; the defensible evidence is the perfect sign inversion across 3 architectures $\times$ 2 splits plus the axis mechanism), and real typing differences between architectures are small (range 0.069–0.088) compared with detection differences (0.007–0.163).

Verdict. PanNuke-CF is a valid *typing stress test*—it ranks architectures’ typing robustness consistently with real transfer—but it is *not* a transfer surrogate: used as one, it inverts the ranking, because its blind spot (detection) is exactly where real transfer differences live. Appearance-only counterfactuals inherit this limitation by construction; probing detection robustness requires interventions on *layout* (density, clutter, artefact), not appearance.

7 Discussion

Why all three pillars went negative, together. The pillars shared one implicit hypothesis—that Dead performance is limited by information (language priors) or data (synthetic examples) rather than by the detection architecture. The diagnosis says otherwise: the foreground head fails to fire on 83% of the missed Dead nuclei (it activates in only 17% of them) while the type branch already “sees” them, and neither language-side fusion, post-hoc recovery, loss re-weighting (3 seeds), real-appearance transplant, nor in-context synthesis with self-consistent labels moved detection. The deficit is consistent with the resolution/architecture of the NP head at $40\times$, a hypothesis we have not yet tested architecturally (e.g. higher-resolution foreground supervision) and which is the main open direction.

Evaluation methodology. Three lessons, each with evidence: (1) seed noise on PanNuke is class-dependent (mPQ 0.002 vs Dead 0.007) and validation-fold Dead signals can point opposite to test (CP1: $+0.024$ vs -0.031)—single-seed, single-fold Dead claims are uninterpretable; (2) released evaluation recipes can embed test-side tuning worth ~ 0.02 mPQ (our HoVer-NeXt-T decode study), larger than typical leaderboard gaps; (3) generated evaluation data is systematically *easier* to segment but *less type-consistent* than real data, so counterfactual benchmarks must be scored arm-vs-control and with strict mPQ alongside mPQ.

Dead as a cross-domain phenomenon. Dead PQ collapsing to 0.002–0.033 on PUMA for all three architectures (in domain 0.13–0.18)—despite Dead being the most text-recognisable class for CONCH and the class our in-domain models type at 0.69 accuracy when detected—suggests the in-domain Dead cue (dark pyknotic chromatin at $40\times$) does not transfer to apoptotic bodies in melanoma H&E. Any Dead-specific contribution must be validated cross-domain, not only on PanNuke folds.

8 Status and Remaining Work

All pre-registered experiments are complete: baselines reproduced (course requirements (a), (b) done), error decomposition and class $\times$ tissue analyses done (c), pillars A/B closed with negative verdicts, pillar C closed with a mechanistic evaluation result. Remaining for the final report: per-tissue $\times$ class heatmaps and error-decomposition figures for all three architectures, a compacted main table set, optional additional external sets (MoNuSeg, NuInsSeg; bPQ/AJI only), and copy-editing. Optional open directions (not pre-registered): architecture-side attack on the Dead detection deficit, and layout-intervention counterfactuals to probe the detection axis that appearance-only CF cannot.

References

- [1] J. Gamper et al., “PanNuke Dataset Name, Instance Segmentation and Classification in Hematoxylin & Eosin Stained Histology Images,” *Medical Image Analysis*, 2020.
- [2] S. Graham et al., “HoVer-Net: Simultaneous Segmentation and Classification of Nuclei in Multi-Tissue Histology Images,” *Medical Image Analysis*, 2019.
- [3] J. S. Horta Malo et al., “CellViT: Vision Transformers for Precise Cell Instance Segmentation,” *Medical Image Analysis*, 2024. arXiv:2306.15350.

- [4] J. S. Horta Malo et al., “CellViT++,” arXiv:2501.05269, 2025.
- [5] S. D. Bhattacharjee et al., “HoVer-NeXt: Exploring Reliable and Efficient Lifted ViT Baselines for Nuclei Segmentation,” *MIDL*, 2024.
- [6] Y. Zhou et al., “CPP-Net: A Unified Framework for Nuclei Instance Segmentation and Classification,” arXiv:2102.06867, 2021.
- [7] S. Graham et al., “PromptNucSeg: A Text-Driven Nuclei Instance Segmentation Method,” *ECCV*, 2024. arXiv:2311.15939.
- [8] R. Lu et al., “A Visual-Language Foundation Model for Computational Pathology (CONCH),” *Nature Medicine*, 2024. arXiv:2307.12914.
- [9] “Mind the Gap: Meta-Analysis of Foundation Models for Pathology Image Analysis,” arXiv:2502.02471, 2025.
- [10] M. Le et al., “PixCell: A Generative Model for Digital Pathology Slides,” arXiv:2506.05127, 2025.
- [11] “NucleiMix: Diffusion-Based Rare-Nuclei Synthesis,” arXiv:2410.16671, 2024.
- [12] “Counterfactual Stress Testing of Medical Imaging Models,” arXiv:2605.10894, 2026.

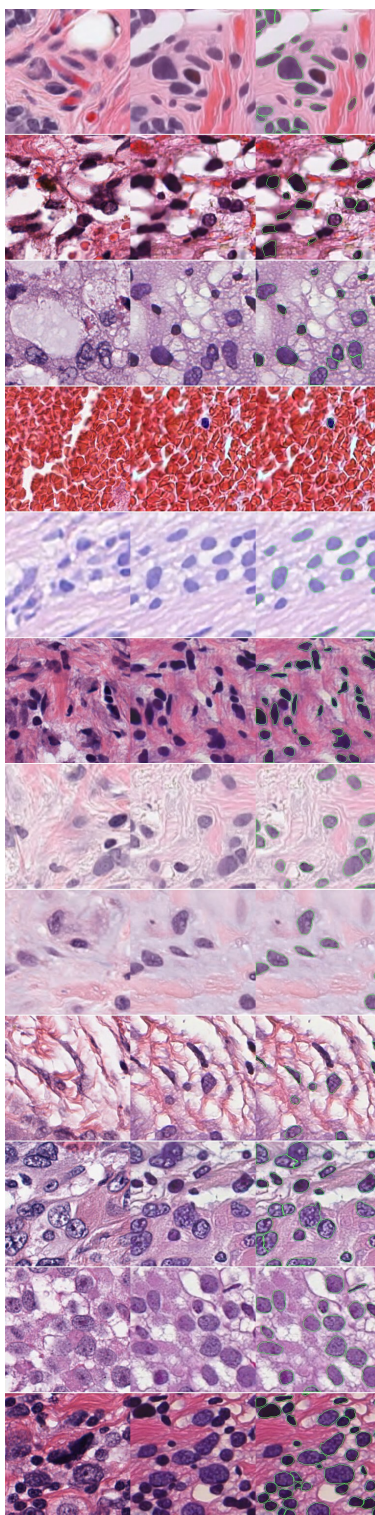


Figure 3: SYN-v2 synthesis examples (real | generated | labels): with paired context + Reinhard the appearance passes review, but the generator’s binary-mask conditioning cannot express class-targeted appearance (no generated object is readable as Dead).

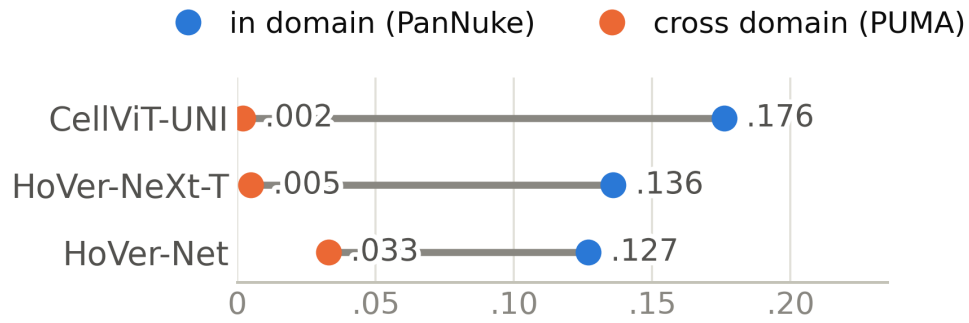


Figure 4: Dead PQ per architecture, in domain (PanNuke, 3-split mean) vs. zero-shot on PUMA: the class nearly vanishes cross-domain for every architecture (0.002–0.033 vs. 0.127–0.176 in domain).

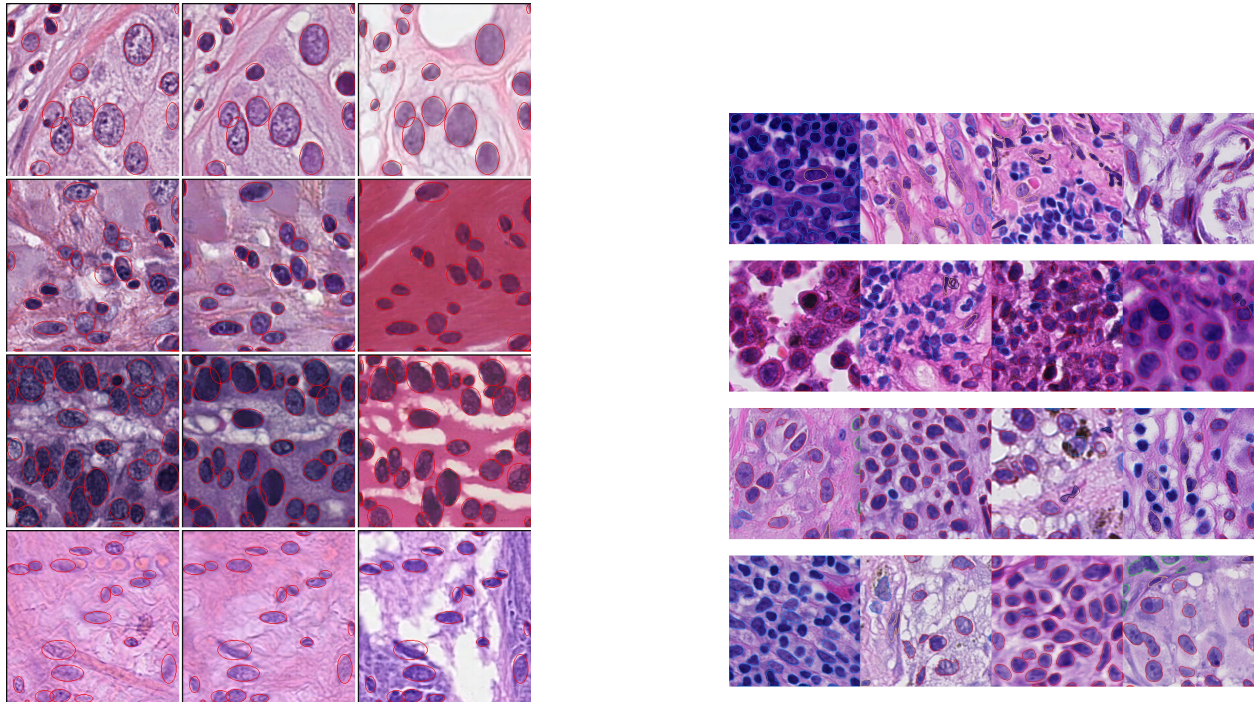


Figure 5: Left: PanNuke-CF arms (same fold-3 GT labels re-rendered under context/stain interventions). Right: PUMA ground truth (colour-coded classes; black outlines = apoptotic Dead), the external set that supplies cross-domain Dead labels.

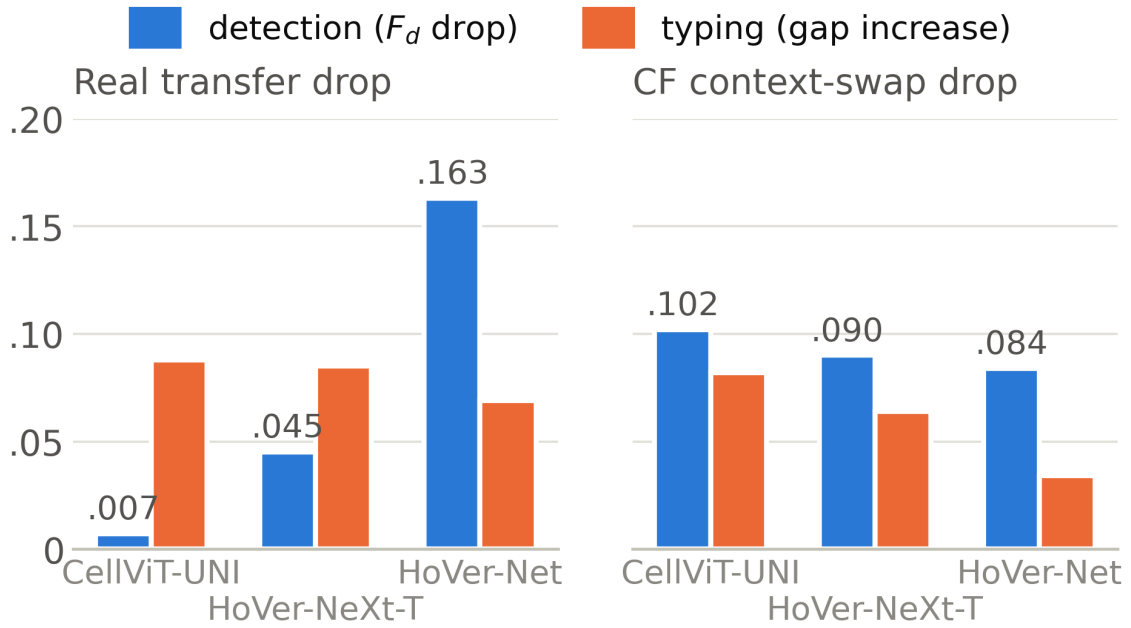


Figure 6: Axis decomposition (architecture means; Table 6): real transfer separates architectures on the detection axis (F_d drop, left), whereas CF context-swap stress is homogeneous in detection and orders only the typing axis (right).